UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2007 question paper

5054 PHYSICS

5054/02 Paper 2 (Theory), maximum raw mark 75

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1 unit penalty per question, no sig. fig penalty **throughout paper**.

Section A

1 **(a)** parachute opens or speed drops from (50 to 5 m/s) **or** decelerates (**e.g.** uniformly) **and** lands/hits ground **or** speed becomes 0 or stops (**e.g.** decelerates) B1

(b) accelerates **or** speed increases (**not** increasing acceleration) B1
acceleration decreases (to 0) **or** speed becomes constant B1

(c) forces balance/cancel **or** no resultant **or** equal and opposite (**not just** forces equal) B1
weight/gravity **and** air resistance/drag mentioned (**not** upthrust/friction) B1

(d) (d =) st **or** s=d/t or any speed x any time **or** area under graph C1
150 m A1 [7]

2 **(a)** **(i)** take reading of liquid before rock placed in **or** pour in a known/specified volume **or** fill eureka can to spout/overflowing B1
take reading with rock and subtract **or** add rock **and** measure overflow B1

(ii) will not fit in **or** volume too large B1

(b) (d =) m/v **or** 101/22 C1
4.59 g/cm^3 A1

(c) C B1
mass/volume **or** density different **or** mass not proportional to volume B1 [7]

3 **(a)** **(i)** geothermal B1

(ii) will not run out **or** infinite **or** being replaced (**not** can be used again/recycled) B1

(b) **(i)** (E =) mcT **or** 1000 x 4200 x 80 **or** whole equation rearranged C1
3.36x 10^8 J A1

(ii) (E=) mL **or** 100 x 2.3 x 10^6 **or** whole equation rearranged C1
2.3x10^8 J A1 [6]

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4	**(a)**	good absorber (**not** good absorber and emitter)		(**not** attracts)	B1	
		radiation **or** infra red		(**not** heat)	B1	
	(b)	hot water rises		(**not** heat rises)	B1	
		by convection (currents) **or** density explanation			B1	
	(c)	**(i)**	reduce/avoid/prevent loss of heat		B1	
		(ii)	cover/wrap in lagging/any sensible material (**not** wood/insulation, **acc.** plastic tank)		B1	[6]
5	**(a)**	**(i)**	atoms vibrate/move back and forth/to and fro	(**accept** particles/molecules)	M1	
			atoms hit neighbours **or** pass on heat/energy to neighbour	(**not** vibrations)	A1	
		(ii)	atoms take up more space/further apart/larger vibrations	(**not** atoms larger)	B1	
	(b)	atoms move throughout (liquid) or not in fixed places or arrangement irregular **or** broken bonds (**e.g.** atoms move faster)			B1	
		atoms move at random/further apart (**e.g.** fixed volume/variable container shape etc.)			B1	[5]
6	**(a)**	cone/molecules vibrate			B1	
		molecules (vibrate) longitudinally/back and forward (in direction of sound) **or** compressions **and** rarefactions mentioned (**e.g.** longitudinal waves)			B1	
	(b)	**(i)**	a number from 18,000 to 22,000 Hz		B1	
		(ii)	(v =) f λ algebraic **or** numerical using 20 Hz **or** candidate's **(i)**		C1	
			17 m		A1	[5]

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7 **(a)** **(i)** steel B1

(ii) rod inside (coil) with current on (at some stage) B1

(b) **(i)** (soft) iron accept Mumetal or any other soft magnetic material B1

(ii) **all** lines directly join from left to right **and** top line goes down **and** bottom line up M1
no lines inside box **and** no lines cross/touch A1 [5]

8 **EITHER**
any regular wave drawn (at least **one** complete wave) B1
amplitude 2 squares B1
time for 1 wave 0.04 (s) or f=1/T seen C1
2 complete waves drawn in 8 squares A1
OR
(a) water conducts/completes (LH) circuit B1
(small) current into (base of) transistor **or** V_{BE} > 0.6 V B1
switches transistor on **or** (large) current from collector to emitter **or** in lamp B1
(lamp switches on alone 0)

(b) any sensible suggestion, e.g., warning of rain (**not** water level for the blind,
not automatic pump/windscreen wipers etc.) B1 [4]

Section B

9 **(a)** **(i)** (acc =) (v-u)/t C1
14/3 C1
4.7 m/s^2 (penalise halving to 2.35 m/s^2, accept 2 or more sig figs **not** fractions) A1

(ii) F = ma or 5 x **(i)** C1
23 N (penalise **second** halving to 5.75 N, ecf **(i)** acc. 2 or more sig figs **not** fractions) A1

(iii) longer time of impact/slows down ball gradually/stops the ball more slowly B1
less acceleration B1 [7]

(b) **(i)** force / area **or** F/A (acc. force on unit area **not** force on an area; N/m^2) B1

(ii) larger area B1
smaller force B1 [3]

(c) **(i)** $P_1V_1 = P_2V_2$ **or** PV = constant C1
1.4 x 10^7 x 600 = P x 30000 **or** 1.4 x 10^7 x 600/30000 C1
280 000 Pa A1

(ii) molecules hit sides (of cylinder) (**not** each other) B1
molecules leave cylinder **or** fewer in cylinder **or** enter air bag B1 [5]

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10 (a) electrical circuit containing cell/source, ammeter/lamp/bell and component under test
or charged gold-leaf electroscope and component **or** other sensible apparatus M1
one correct observation A1
named conductor (any metal/carbon/graphite **accept** water) B1
named insulator (e.g. plastic **accept** paper **and** wood) B1 [4]

(b) (i) voltage/current **or** V/I **not** volts/amps B1

(ii) resistance increases at higher p.d. (**not** resistance increases) B1

(iii) (filament) lamp/bulb or **PTC** thermisitor (**not** metal conductor) B1

(iv) temperature changes C1
higher current/voltage produces higher temperatures A1 [5]

(c) (i) 1.0A **both** for A_1 and A_4 B1

(ii) (V=) IR in any form or 20 x 0.4 C1
8(.0) V A1

(iii) 8 V or same as **(ii)** B1

(iv) **(ii)** / 0.6 C1
13 Ω (accept 2 or more sig figs **or** recurring decimal **not** fractions) A1 [6]

11 (a) (i) (as it enters) bends towards normal B1
(as it leaves) bends away from the normal B1

(ii) speed **and** wavelength change C1
speed **and** wavelength decrease A1
frequency unaltered B1

(iii) sin(i)/sin(r) C1
sin 40°/sin 25° C1
1.5(2) (penalise °, accept 2 or more sig figs; **1.5 alone** with no working B1) A1 [8]

(b) Mark (i) and (ii) separately unless specifically referred to (i) in (ii)

(i) **Words:** distance between (principal) focus/focal point (**not** F) and lens M1
centre of lens A1
Diagram: F/(principal) focus/focal point marked **and** lens marked/curved faces/triangles
and correct arrow of some sort M1
f/FL/fl/focal length marked **and** arrow from centre of lens to F A1

(ii) diagram showing object, lens and **one** correct ray M1
second correct ray M1
correct image shown (½ < h < 1) A1

(iii) smaller / de-magnified / e.c.f **(ii)** B1
upside down B1

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